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① Computer Vision is the branch of Artificial Intelligence which analyze, interpret, understand, recognize the digital image and videos.

a] Objectives of computer vision:

The main objective of the computer Vision is to analyze, image preprocessing, image Recognition, object detection and to understand what the image is by decision making.

b] Two real world examples:

Autonomous Vehicles: Vehicles stops when they detects the pedastrians.

Medical Fields: Taking X-ray of the body to detect what the disease is there.

② Key Features of computer vision:

The computer Vision role is to analyze the image by image preprocessing (i.e) understand by itself. The main key features of computer vision is:

Object Detection

Image Classification

Image Segmentation

Object Recogniyation

Extraction.

a] Feature Extraction:

It is defined as the understanding the useful information in a digital image and video by removing (or) avoiding the unusal objects

b] Object Recognition improves computer vision by classifying the objects in an image using decision making and Machine Learning which improves the efficiency and understand the description of an image.

3] Different Levels of Computer Vision:

To improve the understanding, Analyzing and ~~effect~~ proper Detection of an image computer Vision is classified into three types such as:

* Low Level Image processing.
* Mid Level Image Processing
* High Level Image Processing.

a] Low Level Image processing:

Low Level Image processing is just edits the image which does not understand the features of Image.

The Output will be Image to Image.

Ex: Noise Removal, Image Enhancement, filters etc id-Level Image Processing :

* Mid level is understands the image
* It recognize the objects
* It classifies the image.
* The output will be Image features.

High Level Image Processing :

* High Level both understands the image and detects the objects in an image
* Output will be meaning full information of an image.

4) Digital Image represented in the form of pixels. Where each pixels having Intensity values.

a) A pixel is value two dimensional Matrix in which it having the intensity values. The high pixel value the image resolution will be higher which results in image clarity.

5) Image Resolution affects image quality by while increasing the pixel values the resolution of the Image increases, by decreasing the pixel values the resolution of the picture decreases.

⑤ Image Acquisition:

Image Acquisition caputures the light of the of the object by reflection. The sensor Detects the light and transfers to ~~to~~ digital Signal

a) Role of Sensor:

The Role of Sensor is to capture light which was reflected from the object and to tran it to the lens to convert into digital signal.

b)

* Digital camaras
* Webcam
* CCTV camaras.
* Scanners. etc

6) Sampling and Quantization convert a continuous Image into digital form.

a) Sampling is the process of converting a continuous ~~grid of~~ ~~[illegible]~~ Image by forming grid of pixels. It determines the image spatial resolution

b) It assigns discrete intensity values to sample pixels. It determines the brightness of the Image

8) a) Image Processing is defined as the just editting of the Image in which it the output will be an Image. Functions like:

* Noise Removal
* Image Enhancement.

It ~~[illegible]~~ doesnot have any meaningful information.

Image Processing:

It is just edits the Image and does not have any meaning-ful information.

Computer Vision:

It understand the Image by decision making and machine learning ~~by~~ and have meaning-ful Information.

9) a) Computer Vision helps detecting diseases such as cancer Tumors, memunnea by analyzing the machine Images.

b) It detects signals, Pedastrians, object Detection

10) a) pixels stores brightness & color information which together ~~from~~ the complete Image

b) Sampling determines the no. of pixels while Image resolution affects images Resolution improves accuracy of computer vision system.

(9) The image formation mode. Explains how light from an object passes through a camera lens and forms image on the sensor.

Preppective projection is the process of projecting 3D into 2D images plane while drebiving depth.

Illumnation provides light reflects from objects making them visible; image sensor.